University of Bamberg

Master Project - Machine Listening

(CH-Proj-M)

00 - Topics

Prof. Dr.-Ing. Jakob Abeßer
Professur für Computational Humanities
Fakultät für Wirtschaftsinformatik und Angewandte Informatik
Otto-Friedrich-Universität Bamberg, 96047 Bamberg
E-Mail: jakob.abesser@uni-bamberg.de
Web: https://www.uni-bamberg.de/ch/

T1: Recording and Spatio-Temporal Mapping of Bamberg’s Church Bells

- Description
  - Church bells serve as iconic "soundmarks" that define the acoustic identity and temporal rhythm of urban landscapes
  - This project involves conducting high-quality field recordings of various church bells across Bamberg
  - Students will analyze physical properties like tuning frequencies and materials to create an interactive map visualization

T1: Recording and Spatio-Temporal Mapping of Bamberg’s Church Bells

- Dataset
  - Self-generated field recordings (Bamberg city area)
  - Further explore https://createsoundscape.de/
- Tasks / Research Questions
  - How do material and age correlate with sound profiles?
  - Create a web-based map linking location, sound, and year of construction

T2: Soundscape Analysis and Ecoacoustic Indexing of the ERBA island

- Description
  - The ERBA-Insel represents a unique intersection of urban green space, student life, and industrial history
  - This project uses ecoacoustic indices to quantify the ratio between biological sounds and human-made noise
  - Students will collect data across multiple locations and times to visualize the measurement across the island

T2: Soundscape Analysis and Ecoacoustic Indexing of the ERBA island

- Dataset
  - Self-generated recordings
  - Analyses using the scikit-maad toolbox
- Tasks / Research Questions
  - How do indices like ADI and NDSI vary between weekdays and weekends?
  - How do the indices depend on the geographic location (e.g., proximity to streets)
  - Is spatial interpolation feasible for soundscape mapping?

T3: Bird Activity Detection in Audio

- Description
  - Monitoring avian populations is essential for biodiversity assessment
  - The task focuses on developing robust algorithms to distinguish bird vocalizations from background noise
  - (Binary) bird activity detection is a meaningful first step prior to bird classification

T3: Bird Activity Detection and Species Classification in Audio

- Dataset
  - Warblr10k-raw or BirdVox-70k
- Tasks / Research Questions
  - How robust are different model architectures to noisy conditions?
  - Which non-bird sound sources cause main confusion?
  - What is the optimal window length (e.g., 250ms vs. 2s) for short-duration bird call detection?

T4: Cross-Domain Mosquito Species Classification

- Description
  - Controlling mosquito-borne diseases requires precise monitoring of species distribution in varying acoustic environments
  - This project addresses the challenge of "domain shift" where models trained on one recording setup must perform on another

T4: Cross-Domain Mosquito Species Classification

- Dataset
  - Development dataset from BioDCAse 2026 Challenge (Task 5)
- Tasks / Research Questions
  - How do cross-domain training techniques improve classification accuracy across different recording devices recording locations?

T5: Few-Shot Learning for Rare Bioacoustic Event Detection

- Description
  - In wildlife monitoring, many species are rarely heard, making it difficult to collect large-scale training data
  - The project explores "few-shot" methods where a system learns to detect a sound based on only a few examples
  - The task aligns with the international DCASE community standards for event detection

T5: Few-Shot Learning for Rare Bioacoustic Event Detection

- Dataset
  - [DCASE 2024 Challenge Task 5]
- Tasks / Research Questions
  - Can a model generalize to unseen animal calls using only 5 reference samples?
  - What is the influence of the species call duration / spectral characteristics?

T6: Audio-to-Text Captioning using Large Audio-Language Models (LALMs)

- Description
  - Moving beyond simple classification, sound event captioning describes complex acoustic scenes in natural language
  - This project utilizes state-of-the-art Large Audio-Language Models (LALMs) to "write" descriptions of audio clips

T6: Audio-to-Text Captioning using Large Audio-Language Models (LALMs)

- Dataset
  - Clotho or AudioCaps (typical for these models).
- Tasks / Research Questions
  - How accurately can LALMs describe overlapping sound events compared to traditional tagging?

T7: Acoustic Traffic Monitoring

- Description
  - Acoustic sensors provide a privacy-preserving and cost-effective alternative to cameras for traffic flow analysis
  - The project focuses on classifying vehicle types (bus, car, motorcycle, truck) under varying environmental conditions
  - It specifically investigates the robustness of MEMS vs. high-end microphones

T7: Acoustic Traffic Monitoring

- Dataset
  - IDMT-Traffic dataset
- Tasks / Research Questions
  - Does rain (wet conditions) significantly degrade classification accuracy?
  - How does microphone mismatch impact performance?

T8: Music Genre Classification

- Description
  - Organizing vast digital music libraries requires efficient, automated metadata generation
  - This project involves extracting rhythmic and spectral features to categorize tracks into high-level genres

T8: Music Genre Classification

- Dataset
  - FMA Small dataset (8 genres)
- Tasks / Research Questions
  - Which acoustic features (e.g., MFCCs, chroma) are most indicative of specific genres like Jazz vs. Pop?
  - What are most common confusions among music genres?

T9: Music Chord Recognition

- Description
  - Chords are essential for harmonic analysis of music recordings
  - Chord types range from simple 3-voiced chords (major/minor) to more complex chords (e.g. major-seventh)
  - Chord recognition algorithms must be robust against diverse instrumentation (e.g., piano vs. distorted guitar) and varied recording qualities

T9: Music Chord Recognition

- Dataset
  - IDMT SMT Chords dataset and RWC Popular Music Database.
- Tasks / Research Questions
  - How does instrument timbre (e.g., piano vs. distorted guitar) affect the accuracy of chord detection?
  - Which chords are commonly confused and what is the musicological explanation?

T10: Industrial Sound Analysis (ISA)

- Description
  - Predictive maintenance uses sound to detect machine failures or anomalies before they lead to costly downtime
  - This project explores specific industrial use cases such as leak detection and engine state monitoring
  - Students will apply signal processing to identify subtle deviations in mechanical "fingerprints"

T10: Industrial Sound Analysis (ISA)

- Dataset
  - IDMT-ISA datasets (Electric-Engine, Metal-Balls, or Compressed-Air)
- Tasks / Research Questions
  - Which model architecture provides a good trade-off between performance and size?
  - How does the spectral characteristics of the target sound affect the recognition performance?